Jonathan Drechsel

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Education

2024 – present	**Ph.D. in Computer Science**, University of Passau, Germany
2022 – 2023	**M.Sc. Computer Science**, Dresden University of Technology, Grade 1.1
2021 – 2024	**M.Sc. Mathematics**, Dresden University of Technology, Grade 1.2
2019 – 2021	**B.Sc. Computer Science**, Dresden University of Technology, Grade 1.6
2017 – 2021	**B.Sc. Mathematics**, Dresden University of Technology, Grade 1.8

German grading scale: 1.0 is the best grade, 4.0 the lowest passing grade.

Publications

Drechsel, Jonathan; Herbold, Steffen (2026). The GRADIEND Python Package: An End-to-End System for Gradient-Based Feature Learning. **EMNLP 2026 System Demonstration**

Drechsel, Jonathan; Bytyqi, Erisa; Herbold, Steffen (2026). Understanding or Memorizing? A Case Study of German Definite Articles in Language Models. **ACL 2026 (Oral)**

Drechsel, Jonathan; Herbold, Steffen (2026). GRADIEND: Feature Learning within Neural Networks Exemplified through Biases. **ICLR 2026**

Drechsel, Jonathan; Reusch, Anja; Herbold, Steffen (2025). MAMUT: A Novel Framework for Modifying Mathematical Formulas for the Generation of Specialized Datasets for Language Model Training. **TMLR 2025**

Birkenheuer, Simon; **Drechsel, Jonathan**; Justen, Paul; Pöhlmann, Jimmy; Gonsior, Julius; Reusch, Anja (2023). Sabrina Spellman at SemEval-2023 Task 5: Discover the Shocking Truth Behind this Composite Approach to Clickbait Spoiling! **SemEval 2023**

Woltmann, Lucas; **Drechsel, Jonathan**; Hartmann, Claudio; Lehner, Wolfgang (2022). Ingredient-based Forecast of Sold Dish Portions in Campus Canteen Kitchens. **ICDE Workshops (ICDEW) 2022**

Work Experience

2024 – present	**Research Assistant**, University of Passau Research topics: language models, interpretability, targeted feature learning, math NLP Teaching: assistant for multiple undergraduate and graduate courses
2022 – 2024	**Working Student as Software Developer**, Liofit GmbH
2020 – 2022	**Working Student as Software Developer**, Conimon GmbH

Skills

Programming	Python, Java (very good knowledge); C, C++ (good knowledge)
ML	PyTorch, Transformers
Languages	German (native), English (fluent), French (basic)

Awards

2022 – 2023	Deutschlandstipendium (Germany Scholarship)